FIITJEE

ALL INDIA TEST SERIES

FULL TEST – X

JEE (Advanced)-2019

PAPER – 2

TEST DATE: 05-05-2019

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-D.**

1. ***Section-A (01 – 06, 19 – 24, 37 – 42)*** *contains 18 multiple choice questions which have* ***one or more than one correct*** *answer. Each question carries* ***+4 marks*** *for all correct answer.*
   *Partial Marks : **+3** If all the four options are correct but ONLY three options are chosen.*
   *Partial Marks : **+2** If three or more options are correct but ONLY two options are chosen, both of which are correct options.*
   *Partial Marks : **+1** If two or more options are correct but ONLY one option is chosen and it is a correct option.*
   *Zero Marks : **0** If none of the options is chosen (i.e. the question is unanswered).*
   *Negative Marks : **–2** In all other cases.*

   Section-A (07 – 10, 25 – 28, 43 – 46) *contains 12 questions. Each question has* ***TWO (02)*** *matching lists:* ***LIST-I*** *and* ***LIST-II****.* ***FOUR*** *options are given representing matching of elements from* ***LIST-I*** *and* ***LIST-II****.* ***ONLY ONE*** *of these four options corresponds to a correct matching.*
   For each question, choose the option corresponding to the correct matching.
   Each question has ***only one correct*** *answer and carries* ***+3 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

2. ***Section-D (11 – 18, 29 – 36, 47 – 54)*** *contains 24 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+3 marks*** *for correct answer. There is no negative marking.*

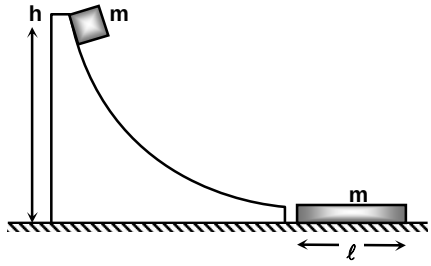
Physics

PART – I

SECTION – A
(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

1. A block of mass m is released from height h from a fixed smooth inclined plane. At the foot plank of mass m is kept on smooth surface. Friction coefficient between plank and block is μ and block just fails to leave the plank of length ℓ. Pick the correct options for the work done by friction on the block and the plank.



(A) $-\frac{mgh}{4}$

(B) $+\frac{mgh}{2}$

(C) $-\frac{3mgh}{4}$

(D) $+\frac{mgh}{4}$

2. A thin hoop of mass 5 kg and radius 10 cm is placed on a horizontal plane. At the initial instant, the hoop is at rest. A small washer of mass 4 kg with zero initial velocity slides from the upper point of the hoop along a smooth groove in the inner surface of the hoop. Speed of the centre of the hoop at the moment when the washer is at a certain point of the hoop, whose radius vector forms an angle 60° with the vertical downward direction is (The friction between the hoop and the plane should be neglected)

(A) $2\sqrt{\frac{1}{3(5+2\sqrt{3})}}$ m/s

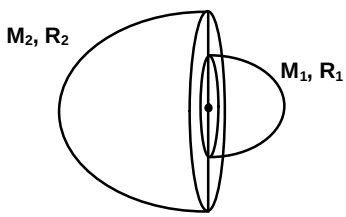
(B) $2\cdot\sqrt{\frac{5-2\sqrt{3}}{21}}$ m/s

(C) $\frac{1}{\sqrt{6}}$ m/s

(D) $\sqrt{6}$ m/s

3. Which of the following is correct about a black body?
   (A) A black body in thermal equilibrium must emit radiation at a rate lower than as it absorbs.
   (B) Objects around room temperature radiate mainly in the infrared region of spectrum.
   (C) At very high temperature (10000 K or more); an iron object is seen blue.
   (D) At temperature 2898 K, the wavelength of maximum intensity is 1 μm.

4. Two hemispherical shells having uniform mass distribution are kept as shown in figure. Force exerted by one hemisphere on the other is 6.67×10^{-11} Newton. Pick the correct options:



(A) Masses and radii are 2 kg, 4 kg and $\sqrt{2}$ m, 2 m respectively.

(B) Masses and radii are 1 kg, 2 kg and $\frac{1}{2}$ m, 1 m respectively.

(C) Masses and radii are 2 kg, 2 kg and 1 m, $\sqrt{2}$ m respectively.

(D) Masses and radii are 1 kg, 1 kg and $\frac{1}{2}$ m, 1 m respectively.

5. Pick the correct options out of the following.
   (A) Coercivity is measure of the reverse field needed to drive the magnetization to zero after being saturated.
   (B) Remanence is measure of the remaining magnetization when the driving field is dropped to zero.
   (C) For ferromagnetic materials, relative permeability is very close to 1.
   (D) Earth has a magnetic field very much like that of a bar magnetic, directed from the geographic south pole to geographic north pole.

6. A ring of thin wire with active resistance R and inductance L rotates with constant angular velocity ω in the external uniform magnetic field perpendicular to the rotation axis. In this process the flux of magnetic induction of external field across the ring varies with time as $\phi = \phi_0 \cos \omega t$. Pick the correct option(s).

   (A) Power developed in the circuit due to rotation is given by $\frac{\phi_0^2\omega^2}{\sqrt{(\omega L)^2 + R^2}}\sin^2 \omega t$.

   (B) The mean mechanical power developed by external forces to maintain rotation is equal to $P = \frac{1}{2}\frac{\omega^2\phi_0^2 R}{(R^2 + \omega^2 L^2)}$

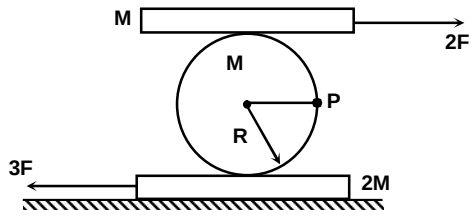
   (C) The mean power developed in the circuit is equal to $P = \frac{1}{2}e_0 I_0 \cos\phi$, where $\tan\phi = \frac{\omega L}{R}$ and I_0 is peak current.

   (D) The mean mechanical power developed by external forces to maintain rotation is equal to $\frac{1}{2}I_0^2 R$, where I_0 is peak current.

(Matching List Type)

This section contains **FOUR** questions. Each question has **TWO** matching lists: **LIST - I** and **LIST - II**. **FOUR** options are given representing matching of elements from **LIST - I** and **LIST - II**. **ONLY ONE** of these four options corresponds to a correct matching.

7. A cylinder is sandwiched between two planks shown in the figure. Two constant horizontal forces 2F and 3F are applied on the planks at t = 0 as shown. There is no slipping at the top and bottom of cylinder. Surface below the plank is smooth.

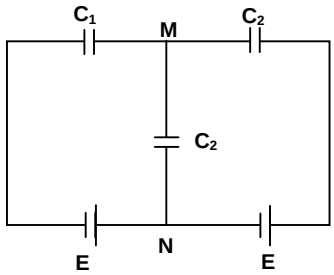


	LIST–I		LIST–II
P.	Acceleration of Centre of mass of cylinder at t = 0	1.	$\frac{20F}{13M} < a < \frac{21F}{13M}$
Q.	Acceleration of top plank at t = 0	2.	$\frac{19F}{13M} < a < \frac{20F}{13M}$
R.	Acceleration of bottom plank at t = 0	3.	$\frac{F}{13M} < a < \frac{2F}{13M}$

S.	Acceleration of point P of cylinder at t = 0	4.	$\frac{17F}{13M} < a < \frac{18F}{13M}$

(A) P → 3; Q → 2; R → 4; S → 1
(B) P → 2; Q → 4; R → 3; S → 1
(C) P → 3; Q → 1; R → 4; S → 2
(D) P → 2; Q → 4; R → 1; S → 3

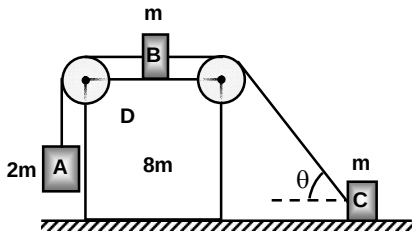
8. In the given circuit two batteries of emf E and E and 3 capacitors of capacitance C_1, C_2 and C_2 are connected. Below is a List – II for the different set of values of potential difference between M and N and value of E with $\frac{C_1}{C_2}$ are given in List-I.



LIST–I		LIST–II	
P.	$\frac{C_1}{C_2} = 2$; E = 100 Volt	1.	0 Volt
Q.	$\frac{C_1}{C_2} = 1$; E = 200 Volt	2.	100 Volt
R.	$\frac{C_1}{C_2} = 3$; E = 100 Volt	3.	25 Volt
S.	$\frac{C_1}{C_2} = 4$; E = 200 Volt	4.	40 Volt

(A) P → 3; Q → 2; R → 4; S → 1
(B) P → 3; Q → 1; R → 4; S → 2
(C) P → 4; Q → 3; R → 2; S → 1
(D) P → 4; Q → 1; R → 2; S → 3

9. A block A of mass 2m is hanging by the pulley fixed on a block D of mass 8m. Another block B of mass m is placed on D whereas a third block C of mass m is on the ground passing by a pulley fixed on the block D. Given that m = 1 kg and angle θ shown in the figure is 60°. The system (maintained in equilibrium) is made free at time t = 0. Answer the following questions in MKS.

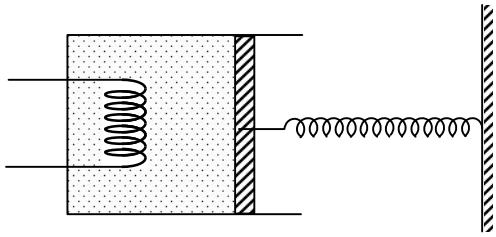


LIST–I		LIST–II	
P.	Acceleration of block D at t = 0 is	1.	$\frac{200}{61}$

Q.	Tension in the slant string attached to C is	2.	$\frac{680}{61}$
R.	Tension in the string joining A and B	3.	$\frac{820}{61}$
S.	Acceleration of block A at t = 0	4.	$\frac{60}{61}$

(A) P → 3; Q → 2; R → 4; S → 1
(B) P → 3; Q → 1; R → 4; S → 2
(C) P → 4; Q → 2; R → 3; S → 1
(D) P → 4; Q → 1; R → 2; S → 3

10. An ideal mono atomic gas is confined by a spring loaded massless piston of cross section 4.0×10^{-3} m^2. Initially the gas is at 400 K and occupies a volume of 4.8×10^{-3} m^3 and the spring is in its relaxed state. The gas is heated by an electric heater until the piston moves out slowly without friction by 0.1 m. Force constant of the spring is 10000 N/m and atmospheric pressure is 1.0×10^5 N/m^2. The cylinder and the piston are thermally insulated.



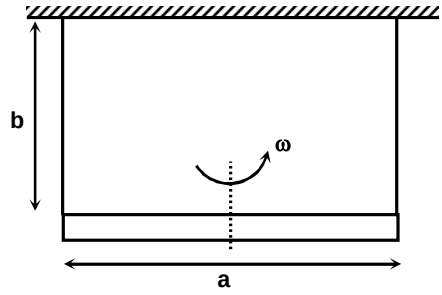
	LIST–I		LIST–II
P.	Heat supplied by the heater is	1.	2100 J
Q.	Additional amount of heat required to compress the spring by additional 0.1 m is	2.	2010 J
R.	Internal energy change of the gas when piston moves 0.1 m from start	3.	2500 J
S.	Internal energy change of the gas when pistons moves 0.2 m from start.	4.	4320 J

(A) P → 4; Q → 1; R → 2; S → 3
(B) P → 2; Q → 3; R → 1; S → 4
(C) P → 3; Q → 2; R → 4; S → 1
(D) P → 1; Q → 3; R → 2; S → 4

SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

11. A rod of mass 1 kg and length 2 m is suspended from the ceiling with the help of two light inextensible cords each of length 1 m so that the rod is horizontal. The rod is given an angular velocity 2 rad/sec about its center vertical axis. Tension in a string immediately after the rod is given angular velocity is 7K Newton. K is



12. An equilateral triangular lamina of mass 1 kg and side length 10 cm is rotating with constant angular velocity 4 rad/sec about an axis passing through centroid O of the triangle and perpendicular to the plane of lamina. The axis is rotating with constant angular velocity 3 rad/sec. Rate of charge of angular momentum of triangular lamina is

13. A rod of length 1 meter has variable coefficient of thermal expansion $\alpha = (1.67 + 1.67x - 1.67x^2) \times 10^{-4}$ (in MKS). The rod is uniform throughout and its Young's modulus is same throughout the rod. Rod is rigidly fixed between two vertical walls. Find the distance x (in MKS) of the point of the rod from corner x = 0 which is not displaced from its position on heating the rod by 1°C. (Given that x is neither 0 nor 1)

14. A light beam of diameter $3\sqrt{3}$ cm in the shape of a hollow cylinder is incident symmetrically around the radius of symmetry on a glass hemisphere of radius 3 cm and refractive indices $\sqrt{3}$ and 1.7 for violet and red colours. Distance between the red and violet colour of light in flat face of hemisphere is (in cm)

15. According to Moseley's Law, the ratio of the slope of graph between $\sqrt{\nu}$ and z (here ν and z are frequency of radiation and atomic number respectively) for K_β and K_α is given by $\frac{n^{\frac{3}{2}}}{\sqrt{2}}$. n is equal to

16. A magnetic field of 3.0×10^{-3} T along z-axis exerts a force $(3\hat{i} + 4\hat{j}) \times 10^{-10}$ N on a particle having a charge 2×10^{-9} C and moving in x-y plane. Here $\hat{i}$ and $\hat{j}$ are unit vectors for x-axis and y-axis respectively. Velocity of particle is given by $(x\hat{i} + y\hat{j})$ m/s. $|(x + y)|$ is equals to

17. A solid sphere of radius 14 cm rolls without slipping in a cylindrical curve of radius 70 cm. The time period of small oscillations is equal to (in sec) (given that g = 10 m/s^2)

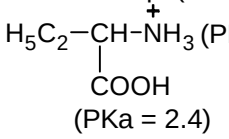
18. A solid sphere of radius 3 cm made of material of bulk modulus 2×10^9 N/m^2 is surrounded by a liquid in a cylindrical container. A massless piston of area 8×10^{-3} m^2 (area of the container is also same) floats on the surface of liquid. A mass of 5 kg is placed on the piston to compress the liquid, change in radius of sphere (in Å) is

Chemistry PART – II

SECTION – A
(One or More than one correct type)

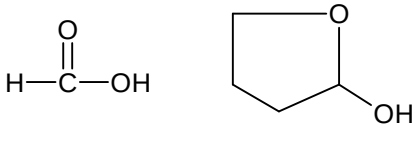
This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

19. What is the pI (iso-electric point) of the following amino acid:

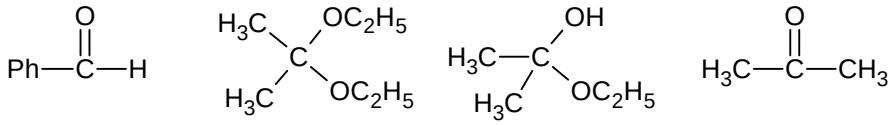


(A) 3.5
(B) 5.9
(C) 2.3
(D) 4.6

20. Which of the following compounds give positive test with Tollen's reagent:



(1) (2) (3) (4)



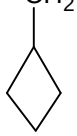
(5) (6) (7) (8)

(A) 1, 2, 3
(B) 3, 4, 5
(C) 6, 7, 8
(D) 1,3, 5

21. Which of these properties is/are correct for Beryllium?
(A) Exhibiting amphoteric nature in their oxides
(B) Forming polymeric chloride by forming (3c, $4e^-$) bond
(C) Forming polymeric hydride by forming (3c, $2e^-$) bond
(D) Forming co-valent halides

22. $P_4 + NaOH(aq) \rightarrow ? + ?$
Select correct statement(s) about the reaction:
(A) One of the product is PH_3
(B) It is disproportionation reaction
(C) One of the product is NaH_2PO_3
(D) This reaction is not given by red phosphorous

23. What is the IUPAC name of the following compound:



(A) Butylcyclobutane
(B) Cyclobutylbutane
(C) Butanecyclobutane
(D) Cyclobutanebutane

24. Which of the following reaction(s) produce(s) Cr_2O_3 along with O_2?
(A) Heating $(NH_4)_2Cr_2O_7$
(B) Heating CrO_3
(C) Heating $K_2Cr_2O_7$
(D) Reaction of Al_2O_3 with Cr

(Matching List Type)

This section contains **FOUR** questions. Each question has **TWO** matching lists: **LIST - I** and **LIST - II**. **FOUR** options are given representing matching of elements from **LIST - I** and **LIST - II**. **ONLY ONE** of these four options corresponds to a correct matching.

25. Match the following List – I with List - II:

List – I		List – II	
P.	$[Ni(CN)_4]^{2-}$	1.	Paramagnetic and number of unpaired electrons(s) is/are same as the free metal ion
Q.	$K_3[Fe(CN)_6]$	2.	Paramagnetic but number of unpaired electron(s) is/are less than the free metal ion
R.	$[NiCl_4]^{2-}$	3.	Free metal ion is paramagnetic but complex is diamagnetic
S.	$[Ni(NH_3)_6]SO_4$	4.	d-orbital(s) is/are involved in hybridization
		5.	Metal in +2 oxidation state
		6.	Metal in +3 oxidation state.

The correct option is
(A) P → 2, 4, 5, Q → 3, 4, 6, R → 1, 4, 5, S → 1, 5
(B) P → 3, 4, 5, Q → 2, 4, 6, R → 1,5, S → 1, 4, 5
(C) P → 1, 5, Q → 3, 4, 6, R → 2, 4, 5, S → 1, 5
(D) P → 2, 4, 5, Q → 1, 4, 6, R → 3, 4, 5, S → 1, 5

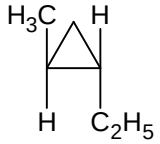
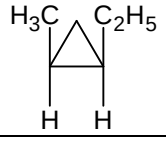
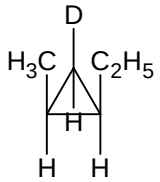
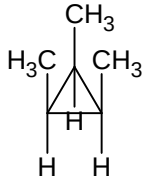
26. Match the following List – I with List - II:

List – I		List – II	
P.	Zn^{2+}	1.	1st group cation
Q.	Ag^{+}	2.	Precipitated by Na_2S
R.	Pb^{2+}	3.	Yellow precipitate with KI
S.	Bi^{3+}	4.	Produce precipitate on dilution with water
		5.	Produce amphoteric hydroxide

The correct option is
(A) P → 2, 5, Q → 1, 2, 3, R → 1, 2, 3, 5, S → 2, 4
(B) P → 3, 4, Q → 1, 4, 5, R → 1, 2, 3, 4, 5, S → 3, 5

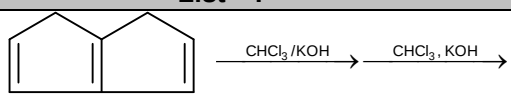
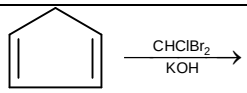
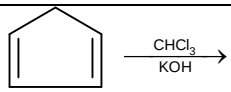
(C) $P \rightarrow 2, 5, Q \rightarrow 1, 4, 5, R \rightarrow 1, 2, 3, 5, S \rightarrow 3, 5$
(D) $P \rightarrow 3, 4, Q \rightarrow 1, 2, 3, R \rightarrow 1, 2, 3, 4, 5, S \rightarrow 2, 4$

27. Match the list:

List – I		List – II	
P.		1.	Chiral
Q.		2.	Achiral
R.		3.	Geometrical isomerism
S.		4.	Containing plane of symmetry
		5.	Compound containing even number of chiral centers

The correct option is
(A) $P \rightarrow 2, 3, 4, Q \rightarrow 1, 3, 5, R \rightarrow 2, 3, 4\ S \rightarrow 1, 3$
(B) $P \rightarrow 1, 3, 5, Q \rightarrow 2, 3, 4, R \rightarrow 1, 3, S \rightarrow 2, 3, 4$
(C) $P \rightarrow 1, 3, 5, Q \rightarrow 1, 3, 5, R \rightarrow 2, 3, 4, S \rightarrow 1, 3$
(D) $P \rightarrow 1, 3, 5, Q \rightarrow 1, 3, 5, R \rightarrow 1, 3, S \rightarrow 2, 3, 4$

28. Match the following List – I with List - II:

List – I		List – II	
P.		1.	Dichloro aromatic compound
Q.		2.	Monochloro aromatic compounds
R.		3.	Monobromo aromatic compound
S.		4.	Carbine formation
		5.	Ring expansion takes place

The correct option is
(A) $P \rightarrow 1, 4, 5, Q \rightarrow 2, 4, 5, R \rightarrow 2, 4, 5, S \rightarrow 2, 4, 5$
(B) $P \rightarrow 2, 4, 5, Q \rightarrow 3, 4, 5, R \rightarrow 3, 4, 5, S \rightarrow 2, 4, 5$
(C) $P \rightarrow 2, 4, 5, Q \rightarrow 3, 4, 5, R \rightarrow 3, 4, 5, S \rightarrow 1, 4, 5$
(D) $P \rightarrow 1, 4, 5, Q \rightarrow 3, 4, 5, R \rightarrow 3, 4, 5, S \rightarrow 1, 4, 5$

SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

29. Calculate the pressure exerted (in atm) by 2 mole of CO_2 gas at 300 K, if volume occupied by CO_2 molecules is negligible. a = 3.592 atm lit^2mol^{-2}.

30. What is the percentage purity of concentrated H_2SO_4 solution (Specific gravity 1.8 g/ml) if 5 ml of this solution is neutralized by 84.6 ml of 1 N NaOH.

31. Calculate the accelerating potential (in volts) that must be imparted to a proton beam to give it an effective wavelength of 0.01 nm.
Given : mass of proton = 1.6×10^{-27} kg
Charge on proton = 1.6×10^{-19} Coulomb
Plank's constant h = 6.6×10^{-34} J/sec

32. An oxide of metal m (at wt = 64) contains 20% oxygen by weight. The oxide was converted into chloride by treatment with HCl and electrolysed. Calculate the amount of metal (in gm) that would be deposited at cathode if a current of 1 ampere was passed for a period of 3.2 hr.

33. The following reaction follows 1st order kinetics at 150^oC:
$A(g) \rightarrow B(g) + C(g)$
On starting with pure A, at the end of 20 min, the total pressure was found to be 300 mm of Hg. After a long time the total pressure of the system was 500 mm of Hg. Calculate the half-life period of the reaction in minutes. (log 1.25 = 0.09691)

34. Calculate the heat of dissociation (in Kcal) for NH_4OH if
$HCl + NH_4OH \rightarrow NH_4Cl + H_2O \quad \Delta H = -12.27$ Kcal
$H^+ + OH^- \rightarrow H_2O \quad \Delta H = -13.7$ Kcal

35. At 470 K partially dissociated vapours of PCl_5 are 60 times as heavy as H_2. Calculate the % dissociation of PCl_5.

36. A gas present in a cylinder fitted with a frictionless piston expands against a constant pressure of 1 atm from a volume of 3 lit. to a volume of 7 lit. In doing so, it absorbs 750 J heat from surroundings. Determine increase in internal energy of the gas. [Take 1 L atm = 101.3 J]

Mathematics PART – III

SECTION – A
(One or More than one correct type)

This section contains **6** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four options is(are) correct.

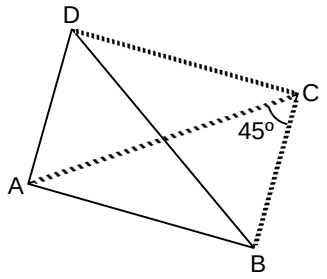
37. Let $f(x) = x^3 - 3x + 1$, then
(A) $f(f(x)) = 0$ has 7 solutions
(B) $f(f(x)) = 0$ has 4 solutions
(C) $f(f(x)) = -1$ has 7 solutions
(D) $f(f(x)) = -1$ has 4 solutions

38. If the system of linear equations $(\cos\theta)x + (\sin\theta)y + \cos\theta = 0$, $(\sin\theta)x + (\cos\theta)y + \sin\theta = 0$, $(\cos\theta)x + (\sin\theta)y - \cos\theta = 0$ is consistent, then the possible values of $\theta \in [0, 2\pi]$ is/are
(A) $\frac{\pi}{2}$
(B) $\frac{3\pi}{4}$
(C) $\frac{3\pi}{2}$
(D) $\frac{7\pi}{4}$

39. Suppose $x, y \in (-2, 2)$ and $xy = -1$ then the least value of $u = \frac{4}{4-x^2} + \frac{9}{9-y^2}$ is $\frac{a}{b}$ (where a and b are coprime), then
(A) $a + b = 17$
(B) $a + b = 19$
(C) $a - b = 7$
(D) $a - b = 5$

40. As shown in the diagram, the volume of tetrahedron DABC is $\frac{1}{6}$.

also $\angle ACB = 45^o$ and $AD + BC + \frac{AC}{\sqrt{2}} = 3$, then



(A) $CD = \sqrt{3}$
(B) $AD = 1$
(C) $BC = 2$
(D) $AC = 2$

41. The number of natural numbers n such that there exists an integers number x for which $499(1997^n + 1) = x^2 + x$, is/are less than
(A) 2
(B) 3
(C) 4
(D) 5

42. If $\cot^{-1}\left(1^2+\frac{3}{4}\right)+\cot^{-1}\left(2^2+\frac{3}{4}\right)+\cot^{-1}\left(3^2+\frac{3}{4}\right)+.....\infty$ is equal to $\tan^{-1}$ k, then k is

(A) prime
(B) even
(C) multiple of three
(D) odd

(Matching List Type)

This section contains **FOUR** questions. Each question has **TWO** matching lists: **LIST - I** and **LIST - II**. **FOUR** options are given representing matching of elements from **LIST - I** and **LIST - II**. **ONLY ONE** of these four options corresponds to a correct matching.

43. Match the following List-I with List-II

LIST – I (Number of solution of)		LIST - II	
P.	$\lvert\sin x\rvert=\frac{\lvert x\rvert}{2}$ is	1.	1
Q.	$e^{-x}+e^{-x^3}=x$ is	2.	2
R.	$\sin^5 x+\cos^3 x=1$ in $(0, 2\pi]$ is	3.	4
S.	$\lvert\cos x\rvert=2\lvert\lvert x\rvert-1\rvert$ in $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$ is	4.	3

The correct option is:
(A) P → 4; Q → 1; R → 3; S → 2
(B) P → 4; Q → 3; R → 1; S → 2
(C) P → 4; Q → 1; R → 2; S → 3
(D) P → 3; Q → 1; R → 4; S → 2

44. Let f: $R^+ \to R$ be an increasing function such that xf(x) + 2 > 0 and $f(x)\cdot f\left(\frac{xf(x)+4}{x}\right)=1$. Match the following List-I with List-II

LIST - I		LIST - II	
P.	The value of [f′(1)] is (where [.] denotes the greatest integer function)	1.	2
Q.	Number of points of inflection of f(x) are	2.	1
R.	The value of [1 – f(1)] is (where [.] denotes the greatest integer function)	3.	0
S.	Number of solutions of sin x = f(8) in the interval $[-\pi, 2\pi]$ are	4.	4

The correct option is:
(A) P → 2; Q → 3; R → 1; S → 4
(B) P → 4; Q → 3; R → 1; S → 2
(C) P → 2; Q → 3; R → 4; S → 1
(D) P → 2; Q → 1; R → 3; S → 4

45. Consider two circles $S_1 \equiv (x-3)^2 + (y-1)^2 = 9$ and $S_2 = x^2 + (y-1)^2 = 9$.
Match the following List-I with List-II

LIST – I		LIST - II	
P.	Length of common tangent to $S_1 = 0$ and $S_2 = 0$ is	1.	9
Q.	Length of common chord to $S_1 = 0$ and $S_2 = 0$ is	2.	$3\sqrt{3}$
R.	Maximum distance between two points, one on $S_1 = 0$ and other on $S_2 = 0$, is	3.	6
S.	Angle between $S_1 = 0$ and $S_2 = 0$ is $\lambda\pi$, then 9λ is	4.	3

The correct option is:
(A) P → 2; Q → 4; R → 3; S → 1
(B) P → 4; Q → 2; R → 1; S → 3
(C) P → 4; Q → 3; R → 2; S → 1
(D) P → 4; Q → 3; R → 1; S → 2

46. Match the following List-I with List-II

LIST - I		LIST - II	
P.	$\int_0^{\pi/2} \sin^5 x\,dx$ is	1.	$\frac{3\pi-8}{12}$
Q.	$\int_0^{\pi/4} \tan^4 x\,dx$ is	2.	$\frac{8}{15}$
R.	$\int_0^{1} x(1-x)^5\,dx$ is	3.	$\frac{1}{120}$
S.	$\int_0^{\pi/2} \sin^7 x\cos^5 x\,dx$ is	4.	$\frac{1}{56}$

The correct option is:
(A) P → 1; Q → 2; R → 3; S → 4
(B) P → 2; Q → 1; R → 4; S → 3
(C) P → 4; Q → 3; R → 2; S → 1
(D) P → 3; Q → 4; R → 1; S → 2

SECTION – D
(Numerical Answer Type)

This section contains **EIGHT** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. xxxxx.xx).

47. Let f(x) and g(x) are polynomial of degree 4 such that $g(\alpha) = g'(\alpha) = g''(\alpha) = 0$. If $\lim_{x\to\alpha} \frac{f(x)}{g(x)} = 0$, then number of different real solutions of equation $f(x)g'(x) + g(x)f'(x) = 0$ is equal to

48. Consider a cube whose faces are given by $x + y + z = 3\sqrt{3}$, $x + y + z = 2\sqrt{3}$, $4x - 5y + z = \sqrt{42}$, $4x - 5y + z = 2\sqrt{42}$, $2x + y - 3z = \sqrt{14}$ and $2x + y - 3z = 2\sqrt{14}$ & a triangle whose vertices are (2, 1, 3), (1, 1, 1), (3, 1, 0), then the number of point of intersection of cube and triangle is

49. Let $A = \left[a_{ij}\right]_{3\times3}$ is a square matrix of order 3 such that adj $A = \begin{bmatrix} 5 & -1 & -7 \\ -1 & -7 & 5 \\ -7 & 5 & -1 \end{bmatrix}$ and det(A) < 0, then $\frac{1}{3}\sum_{i=1}^{3}\sum_{j=1}^{3} a_{ij}$ is equal to

50. If the value of $\sin 1^o \sin 3^o \sin 5^o \ldots\ldots \sin 179^o = \frac{1}{2^{\lambda}}$, then $(\lambda - 82)$ is equal to

51. A_1 and A_2 are the vertices of the conic $C_1 : 4(x-3)^2 + 9(y-2)^2 - 36 = 0$ and point P is moving in the plane such that $\left|PA_1 - PA_2\right| = 3\sqrt{2}$, then the locus of P is another conic C_2. If D_1 denotes the distance between foci of conic C_2. D_2 denotes the product of perpendicular distances from the points A_1, A_2 upon any tangent drawn to the conic C_2 and D_3 denotes length of tangent drawn from any point on auxiliary circle of conic C_1 to the auxiliary circle of the conic C_2, then $\left(\frac{D_1D_2}{D_3^2}\right)^2$ is equal to

52. If (h, k) be a point on the axis of parabola $3y^2 + 4y - 6x + 8 = 0$ from which three distinct real normals could be drawn to the parabola, then $[h]_{least}$ is equal to (where [.] denotes the greatest integer function)

53. If N denotes the sum of all values of x satisfying equation $\left[\frac{x}{2}\right] - \left[\frac{x}{3}\right] = \frac{x}{7}$, then $\left|\frac{N}{7}\right|$ is equal to (where [.] denotes the greatest integer function)

54. For any acute angle ΔABC, let M denotes the maximum value of $\frac{\sin A}{A} + \frac{\sin B}{B} + \frac{\sin C}{C}$, then $\left[\frac{2\pi M}{3}\right]$ is equal to (where [.] denotes the greatest integer function)